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Araştırma Alanları

Tekstil Malzemeleri ve Prosesleri: Yüksek performanslı lifler ve Sürdürülebilir alternatif doğal lifler
Proses Tasarımı ve Analizi: Yüksek performanslı liflerin yeni nesil çevreci yöntemlerle boyanması, Sürdürülebilir terbiye teknolojileri, doğal boya/yardımcı kimyasal alternatifleri
Odak Alanları: Sürdürülebilir ve ileri tekstil malzemeleri, ekolojik taşıyıcılar, boyama proseslerinin optimizasyonu ve çevresel etki performansı

Mevcut Kurum

Doktor Öğretim Üyesi
Pamukkale Üniversitesi Mühendislik Fakültesi
Tekstil Mühendisliği Bölümü, Tekstil Bilimleri ABD, Denizli, Türkiye — Şubat, 2026 - Devam

Araştırma Görevlisi, Dr,
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Tekstil Mühendisliği Bölümü, Tekstil Bilimleri ABD, Denizli, Türkiye — Şubat, 2016 - Şubat, 2026

Eğitim Bilgileri

Doktora, Tekstil Mühendisliği, Pamukkale Üniversitesi, Denizli, Türkiye — Ağustos, 2024
Doktora tezi: "Polieterimid liflerinin boyanma özelliklerinin araştırılması"

Yüksek Lisans, Tekstil Mühendisliği, Pamukkale Üniversitesi, Denizli, Türkiye — Ocak, 2017
Yüksek Lisans Tezi: "Ananas liflerinin ön terbiyesinin araştırılması"

Lisans, Tekstil Mühendisliği, Pamukkale Üniversitesi, Denizli, Türkiye — Haziran, 2010

Uluslararası Eğitim ve Akademik Deneyim

Erasmus Öğrenim Hareketliliği
Ev Sahibi Kurum: Universitat Politecnica de Valencia, Valencia, İspanya — Eylül 2008-Haziran 2009

Uluslararası Akademik Ziyaret / Araştırma Deneyimi
Ev Sahibi Kurum: Kyoto Institute of Technology, Kyoto, Japonya — Temmuz,2016- Ağustos,2016

Desteklenen Projeler

1. Poliester Liflerinin Süperkritik Karbondioksit Ortamında Boyanmasında Kullanılacak Ekolojik Difüzyon Hızlandırıcı Madde Geliştirilmesi ve Prosesin Plazma Tekniği ile Desteklenmesi — 2026 – Devam ediyor

2.	Colorsense - Boyali Kumaş Renk Kalite Kontrolü Süreci İyileştirilmesi İçin Yapay Zeka Destekli Karar Sistemi Tasarimi, Projedeki Görevi: Araştırmacı/uzman, TUBITAK 1505, 2025-2027.	2025 – Devam ediyor
3.	Denizli'de yaygın olarak üretilen bitkilerin polyester liflerinin renklendirilmesinde doğal difüzyon arttırıcı olarak kullanımı, Proje yürütücüsü, Başlangıç Seviyesi Projesi, Pamukkale Üniversitesi, 2025BSP006.	2025 – Devam ediyor
4.	Polieterimid liflerinin boyanma özelliklerinin araştırılması, Projedeki görevi: Araştırmacı, Doktora Tez Projesi, Tamamlandı, 2024.	2018 - 2024
5.	Ananas liflerinin ön terbiyesinin araştırılması, Projedeki görevi: Araştırmacı, Yüksek Lisans Tez Projesi, Tamamlandı, 2016	2015 – 2016

Yayınlar

SCI / SCI-Expanded makaleler

1. Deniz, N. G., Iscan, A., Sayil, C., Avinc, O., **Kalayci, E.** "Naphthoquinone disperse dyes and their dyeing application to polyethylene terephthalate fabrics", The Journal of The Textile Institute, 1–15, (2023). https://doi.org/10.1080/00405000.2023.2271667
2. **Kalayci, E.,** Avinc, O., "The Disperse Dyeing of Polyetherimide Fibers Using Orthogonal Array (Taguchi) Design", Fibers and Polymers, 25 (8), 3045-3060, (2024a). https://doi.org/10.1007/s12221-024-00641-1
3. **Kalayci, E.,** Avinc, O., "Assessing the Washing Fastness and Environmental Impact of Various Reduction Clearing Processes on High-Performance Polyetherimide (PEI) Fabrics", Fibers and Polymers, 25 (6), 2359-2374, (2024b). https://doi.org/10.1007/s12221-024-00580-x
4. **Kalayci, E**., Avinc, O. Advancing Sustainable Textile Dyeing: Investigating Natural Oils as Natural Carriers for Microwave-Assisted Dyeing of PEI Fibers with Natural Turmeric Dye. *Fibers Polym* (2025). https://doi.org/10.1007/s12221-025-01212-8
5. **Kalayci, E**., Avinc, O. & Eren, H.A. Bleaching Processes of Pineapple Fibers Assisted with Sustainable Technologies: Microwave, Ultrasound and Ozonation. *Fibers Polymers* **26**, 5173–5186 (2025). https://doi.org/10.1007/s12221-025-01155-0

Uluslararası Hakemli Dergilerde Yayınlanan Makaleler (Alan İndeksli Dergiler)

1. **Kalayci, E**., Avinc, O., Yavas, A., "Yüksek Performanslı PBO (Poli-p-penilenbenzobisoksazol Lifleri", Tekstil ve Mühendis, 22 (98): 62-83, (2015). https://doi.org/10.7216/130075992015229807
2. **Kalayci, E**., Avinc, O., Yavas, A., "Polibenzimidazol (PBI) Lifleri", Tekstil ve Mühendis, 21 (96): 51-67, (2014). https://doi.org/10.7216/130075992014219606

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1. **Kalayci, E**., Ala, D.M., Topcu, H., Sato, T., İkiz, Y., The Effects of Repeated Laundering and Structural Parameters on the Terry Fabric Preference of Japanese Consumers and Comparison with Turkish Consumers. Cukurova University, Journal of the Faculty of Engineering, 38(3), 695-704, (2023). https://doi.org/10.21605/cukurovaumfd.1377727
2. Turkoglu, K.B., **Kalayci, E**., Avinc, O., Yavas, A., " Oleofilik Buoyans Özellikli Kapok Lifleri ve Yenilikçi Yaklaşımlar ", Düzce Üniversitesi Bilim ve Teknoloji Dergisi, 7, 61-89, (2019). https://doi.org/10.29130/dubited.424120
3. **Kalayci,** E., Avinc, O., Yavas, A., "Yüksek Performanslı Lifler: Polieter eter keton (PEEK) Lifleri", Cumhuriyet Üniversitesi Fen Bilimleri Dergisi, 38 (2): 168-186, (2017). http://dx.doi.org/10.17776/cumuscij.308456)
4. **Kalayci, E**., Avinc, O., Yavas, A., "Tarımsal Atıklardan Elde Edilen Sürdürülebilir Tekstil Lifleri: Ananas Yaprağı Lifleri", Sakarya Üniversitesi Fen Bilimleri Enstitüsü Dergisi, 20 (2): 203-211, (2016). https://doi.org/10.16984/saufenbilder.07521
5. **Kalayci, E**., Avinc, O., Yavas, A., "Yüksek Performanslı Polietilen (HPPE) Lifleri", Marmara Üniversitesi Fen Bilimleri Dergisi, 28 (1): 13-34, (2016). https://doi.org/10.7240/mufbed.51868
6. **Kalayci, E**., Avinc, O., Yavas, A., "Yüksek Performanslı Lifler: Polipridobisimidozal (PIPD) Lifleri", Celal Bayar Üniversitesi Fen Bilimleri Dergisi, 12 (1): 93-101, (2016). http://dx.doi.org/10.18466/cbujos.25094
7. **Kalayci, E**., Avinc, O., Yavas, A., "Yüksek Performanslı Lifler: Polifenilen sülfid (PPS) Lifleri", Düzce Üniversitesi Bilim ve Teknoloji Dergisi, 4 (1): 88-103, (2016).
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Uluslararası kitap ve kitap bölümleri

1. **Kalayci, E**., Avinc, O., Yavas, A., Coskun, S., "Responsible textile design and manufacturing: environmentally conscious material selection", (eds:Alqahtani, A. Y., Kongar, E., Pochampally, K. K., Gupta, S. M.), Responsible Manufacturing: Issues Pertaining to Sustainability, Taylor & Francis, (2019).
2. **Kalayci, E**., Avinc, O., Turkoglu, K. B., "Importance of Asclepias Syriaca (Milkweed) Fibers in Sustainable Fashion and Textile Industry and Its Potential End-Uses", (ed:Muthu, S. S.), Sustainable Approaches in Textiles and Fashion: Fibres, Raw Materials and Product Development, Singapore: Springer Nature Singapore, 1-21, (2022).
3. **Kalayci, E**., Avinc, O., Yavas, A., "An Alternative Fiber Source in Sustainable Textile and Fashion Design: Cellulosic Akund Fibers", (ed:Muthu, S. S.), Novel Sustainable Raw Material Alternatives for the Textiles and Fashion Industry, Cham: Springer Nature Switzerland, 185-198, (2023).
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6. **Kalayci, E**., Avinc, O., "Life Cycle Assessment (LCA) of Biocomposites and Its Impact", (ed:R. Arun Ramnath, M. R. S., Sergey M. Gorbatyuk, Steffen Fischer, Suchart Siengchin), Biocomposites—Bio-Based Fibers and Polymers From Renewable Resources, Elsevier, (2024).

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1. **Kalayci, E.** (2025, September). *Potential of Denizli-grown aromatic plants in functional textile finishing: From regional flora to functional apparel*. In 6th International Congress on Engineering and Life Science (pp. 289–305). Kyrenia, Turkish Republic of Northern Cyprus. (doi: 10.61326/icelis2025kyrenia)
2. **Kalayci, E.** (2025, September). *Digital transformation in the textile industry: From smart manufacturing to sustainability*. In 6th International Congress on Engineering and Life Science (pp. 306–314). Kyrenia, Turkish Republic of Northern Cyprus. (doi: 10.61326/icelis2025kyrenia)
3. **Kalayci, E**., Yavaş, A., & Avinç, O. O. (2019, May 20–21). *Usage of digital and 3D printing technologies in denim jeans design as an ecological solution*. In Textile Science and Economy. Zrenjanin, Serbia
4. **Kalayci, E.,** Yavaş, A., Avinç, O. O., & Tonguç, T. (2019, May 20–21). *Dyeing performance comparison of soybean fibers with direct dyes with conventional and microwave technologies*. In Textile Science and Economy. Zrenjanin, Serbia.
5. Kalayci, C. B., **Kalayci, E.,** Avinç, O. O., & Yavaş, A. (2019, May 20–21). *Color strength estimation of raffia fibers after different alkali treatments*. In Textile Science and Economy IX. Zrenjanin, Serbia.
6. **Kalayci, E.,** Avinç, O., & Yavaş, A. (2019, May 23–24). *Usage of horse hair as a textile fiber and evaluation of color properties*. In Innovative Solutions for Sustainable Development of Textiles and Leather Industry (pp. 57–62). Oradea, Romania.
7. Kalayci, C. B., Gundogan, M., **Kalayci, E.,** & Avinç, O. (2019, May 23–24). *Color strength estimation of coir fibers bleached with peracetic acid*. In Innovative Solutions for Sustainable Development of Textiles and Leather Industry (pp. 65–70). Oradea, Romania.
8. Yavaş, A., Avinç, O. O., **Kalayci, E.,** & Arık, B. (2017, March). *Color properties of bamboo fabric printed with reactive printing paste containing UV absorber*. In Asia and Africa Science Platform Establishment of Collaboration Research for Neo-Fiber Technology in Asia and Africa+ Seminar Series 10 and Annual Report Meeting for 'Neo-Fiber Technology' Project 2016–2017 (pp. 102–103). Kyoto, Japan.
9. Yavaş, A., **Kalayci, E.,** Avinç, O. O., Arık, B., Çalışkan, A., Börekçi, B. E., & Hacıali, A. (2017, March). *Water-oil repellent treatment of viscose fabric with microwave irradiation*. In Asia and Africa Science Platform Establishment of Collaboration Research for Neo-Fiber Technology in Asia and Africa+ Seminar Series 10 and Annual Report Meeting for 'Neo-Fiber Technology' Project 2016–2017 (pp. 104–105). Kyoto, Japan
10. **Kalayci, E.,** Yavaş, A., Avinç, O., & Arpacı Gündoğan, M. (2017, March). *The effects of pectinase enzyme treatment on the hydrophilicity properties of pineapple fabrics*. In International Conference on Advances in Science and Arts (ICASA'17). Istanbul, Turkey.
11. **Kalayci, E.,** Avinç, O., Yavaş, A., & Arpacı Gündoğan, M. (2017, March). *Peracetic acid bleaching of natural lignocellulosic pineapple fibers*. In International Conference on Advances in Science and Arts (ICASA'17). Istanbul, Turkey.
12. Gündoğan, M., Avinç, O., Yavaş, A., & **Kalayci, E.** (2017, March). *Color properties of coir fiber after different enzyme treatments*. In International Conference on Advances in Science and Arts (ICASA'17). Istanbul, Turkey.
13. **Kalayci, E**., Yavaş, A., Avinç, O., Köroğlu, A. N., & Arpacı Gündoğan, M. (2017, March). *Gamma irradiation and its textile applications*. In International Conference on Advances in Science and Arts (ICASA'17). Istanbul, Turkey.
14. **Kalayci, E.,** Yavaş, A., & Avinç, O. O. (2017, May). *The effects of cellulase and laccase enzyme treatments on pineapple fibers*. In 16th Symposium on Recent Developments in Textile Technology and Chemistry. Bursa, Turkey.

15. **Kalayci, E**., Avinç, O. O., & Yavaş, A. (2016, May). *Futuristic design approaches for new generation textiles*. In Textile Science and Economics VIII. Zrenjanin, Serbia.
16. **Kalayci, E.,** Avinç, O. O., & Yavaş, A. (2016, May). *Textile related extreme sky sports*. In Textile Science and Economics VIII. Zrenjanin, Serbia.
17. Yavaş, A., Ozguney, A. T., Avinç, O. O., & **Kalayci, E.** (2016, May). *The effects of remnant pretreatment chemicals on the reactive printing quality of cotton fabric*. In Textile Science and Economics VIII. Zrenjanin, Serbia.
18. Yavas, A., Avinç, O. O., & **Kalayci, E.** (2016, May). *The effects of different finishing applications on the colorimetric and color fastness properties of reactive dyed cotton fabrics*. In Textile Science and Economics VIII. Zrenjanin, Serbia.
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20. **Kalayci, E.,** Avinç, O. O., & Yavaş, A. (2016, October). *Ethylene tetrafluoroethylene (ETFE) materials for building textiles*. In 16th International Materials Symposium (IMSP). Denizli, Turkey
21. **Kalayci, E.,** Yıldırım, F. F., Avinç, O. O., & Yavaş, A. (2015, May). *Textile fibers used in products floating on the water*. In Textile Science and Economics VII. Zrenjanin, Serbia.
22. **Kalayci, E**., Yıldırım, F. F., Avinç, O. O., & Yavaş, A. (2015, May). *From sea and oceans to the textiles*. In Textile Science and Economics VII. Zrenjanin, Serbia.
23. Kalayci, C. B., & **Kalayci, E.** (2015, May). *Assembly line balancing using an ant algorithm in the garment industry: A case study*. In Textile Science and Economics VII. Zrenjanin, Serbia.
24. Yıldırım, F. F., **Kalayci, E**., Avinç, O. O., & Yavaş, A. (2015, May). *The usage of ultrasonic bath for elastic polyester dyeing*. In Textile Science and Economics VII. Zrenjanin, Serbia.
25. Yıldırım, F. F., **Kalayci, E**., Avinç, O. O., & Yavaş, A. (2015, May). *Color fastness properties of poly(butylene terephthalate) fibers dyed in ultrasonic bath*. In Textile Science and Economics VII. Zrenjanin, Serbia
26. Avinç, O. O., **Kalayci, E**., Yıldırım, F. F., & Yavaş, A. (2015, December). *Biologically inspired some textile natural fibers*. In 2nd Tokyo International Conference on Life Science and Biological Engineering (ILSBE). Tokyo, Japan.
27. **Kalayci, E**., Avinç, O. O., & Yavaş, A. (2014, October). *One of the promising technical textile fiber*. In Textile Science and Economics VI. Zrenjanin, Serbia
28. Avinç, O. O., Yıldırım, F. F., Yavaş, A. & **Kalayci, E**., (2017, May). *3D printing technology and its influences on the textile industry*. In IIER International Conference. Beijing, China. (Ayrıca makale olarak basılmıştır: Avinc, O., Yildirim, F. F., Yavas, A., & Kalayci, E. (2017). 3D printing technology and its influences on the textile industry. *International Journal of Industrial Electronics and Electrical Engineering*, *5*(7), 37. http://iraj.in)
29. **Kalayci, E.,** Yavaş, A., & Avinç, O. (2018, November 22–23). *The effects of different alkali treatments with different temperatures on the colorimetric properties of lignocellulosic raffia fibers*. In International Conference on Recent Advances in Engineering and Technology. Havana, Cuba. (Ayrıca makale olarak basılmıştır: Kalayci, E., Avinc, O., & Yavas, A. (2019). The effects of different alkali treatments with different temperatures on the colorimetric properties of lignocellulosic raffia fibers. *International Journal of Advances in Science Engineering and Technology*, *7*(Special Issue 1), 15. http://iraj.in
30. Yıldırım, F. F., Kahvecioğlu Sarı, H., Avinç, O. O., Yavaş, A., **Kalaycı, E**., & Çörekcioğlu, M. (2019). Evaluation of some physical properties of silk fabrics marble printed with natural colorant. In International Congress on Wool and Luxury Fibres (ICONWOOL 2019) (pp. 130–134). Çorlu, Tekirdağ, Türkiye.
31. Yıldırım, F. F., Kahvecioğlu Sarı, H., Yavaş, A., Avinç, O. O., **Kalaycı, E.,** & Çörekcioğlu, M. (2019). Crease recovery characteristics of the marble printed silk fabrics with synthetic ink before and after repeated washings. In International Congress on Wool and Luxury Fibres (ICONWOOL 2019) (pp. 124–129). Çorlu, Tekirdağ, Türkiye.
32. Yavaş, A., **Kalayci, E.,** Avinç, O. O., & Aras, G. (2017, March). *Microwave energy fixation for discharge on cotton fabric*. In International Conference on Engineering Technology and Innovation (ICETI). Sarajevo, Bosnia and Herzegovina.
33. **Kalayci, E**., Yavaş, A., & Avinç, O. O. (2017, March). *Potassium permanganate bleaching of natural pineapple fibers*. In International Conference on Engineering Technology and Innovation (ICETI). Sarajevo, Bosnia and Herzegovina.

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2. Yıldırım, F. F., **Kalaycı, E.,** Avinç, O. O., & Yavaş, A. (2015). Farklı boyama şartlarının elastik polyesterlerin renk özelliklerine etkisi. In 15. Tekstil Teknolojisi ve Kimyasındaki Son Gelişmelere Sempozyumu. Bursa, Türkiye.
3. Yıldırım, F. F., **Kalaycı, E.,** Avinç, O. O., & Yavaş, A. (2015). Mikrodalga enerjisi kullanılarak boyanmış politrimetilen teraftalat (PTT) liflerinin kolorimetrik özelliklerinin incelenmesi. In 15. Tekstil Teknolojisi ve Kimyasındaki Son Gelişmelere Sempozyumu. Bursa, Türkiye.

Hakemlikler

- Cellulose
- Aquatic Ecology
- Tekstil ve Konfeksiyon
- Coloration Technology
- Chemical Papers
- Tekstil ve Mühendis
- Adıyaman Üniversitesi Mühendislik Bilimleri Dergisi
- Harran Üniversitesi Mühendislik Dergisi

Yabancı Diller

- İngilizce (İleri), İspanyolca (Başlangıç)

Dersler

- 2025-2026 Bahar Dönemi, TENG 455, Mühendislik Etiği
- 2025-2026 Bahar Dönemi, TENG 613, Digital Transformation in Textile and Artificial Intelligence Applications,